IT24100853 - Gayathri M.G.K
Assigned Model: Logistic Regression

1. Individual Role and Contributions

- My main responsibility in the GrainWise project was developing and evaluating the Logistic Regression classification model to distinguish Cammeo and Osmancik rice varieties. In addition to model development, I contributed to exploratory data analysis (EDA), baseline evaluation, shared preprocessing functions, and the final model comparison.
- I worked on the Logistic_Regression.ipynb notebook, performing hyperparameter tuning, cross-validation, feature coefficient analysis, and model stability testing. I also contributed to the shared evaluation pipeline to maintain consistent dataset splitting and evaluation across the team's models.

2. Technical Learning and Key Decisions

- One of my most important learning experiences was understanding how preprocessing affects machine learning performance. We used an 80/20 stratified train-test split and applied standardisation within the modelling pipeline to prevent data leakage during validation.
- For Logistic Regression, I used GridSearchCV to tune regularisation strength, penalty type, and class weighting. The selected configuration used L2 regularisation with C = 3.1623 and the liblinear solver.
- The model achieved 93.07% cross-validation accuracy, 91.85% F1-score, and 97.99% ROC-AUC. These results helped me understand why classification models should be evaluated using several metrics instead of accuracy alone.

3. Challenges and Problem-Solving

- A major challenge was handling highly correlated features, particularly Area and Convex_Area, which could affect the interpretation of Logistic Regression coefficients. I explored alternative feature sets and analyzed model coefficients to understand their influence on predictions.
- I also investigated whether the model's performance depended heavily on the random train-test split. Testing different random seeds produced similar accuracy values, increasing confidence in the stability of the results.

4. Critical Evaluation and Reflection

- Comparing Logistic Regression with SVM, KNN, Random Forest, and baseline models taught me that the highest accuracy does not automatically prove statistical superiority. Although Logistic Regression performed slightly better, the differences were not statistically significant.
- I also learned how classification thresholds influence business decisions. Under the project's assumed 5:1 false-positive-to-false-negative cost ratio, increasing the threshold to 0.805 reduced the held-out test misclassification cost from 204 to 116 units, while reducing accuracy.

5. Personal Growth and Future Improvements

- This project improved my practical understanding of model development, feature selection, validation, interpretation, and collaborative machine learning workflows. Most importantly, I learned to justify technical decisions using experimental evidence rather than assumptions.
- In future projects, I would explore fully nested cross-validation and evaluate the model using rice images collected under different camera conditions to improve its reliability in real-world applications.